Entrance Examination for the Department of Electrical and Electronic Engineering, Graduate School of Engineering, University of Tokyo

Mathematics

9:00 ~ 11:00, Tuesday, August 28, 2007

Instructions

1. Do not open this booklet before permission is given.

2. This booklet contains four problems. If you find missing or badly printed pages, ask the proctor for exchange.

3. Fill in the top parts of all your two answer sheets and scratch paper. Before submitting your answer sheets, make sure that the top parts are correctly filled in.

4. **Answer two problems out of the four.** Your answer to each problem should be written on a separate sheet. You may use the reverse side of the sheet if necessary.

5. The two answer sheets must be submitted at the end of the examination, even if they are blank ones. (The top parts of all the two sheets must be filled in.)

6. You must answer in Japanese or English.

7. This booklet and the scratch paper must be returned at the end of the examination.

8. Problems in Japanese are regarded as official ones. This English translation is provided for the convenience of applicants.

Applicant Number	

Write your applicant number in the above box.

Problem 1

Consider the Bernoulli differential equation

$$(x^2+1)\frac{dy}{dx}+4xy=4axy^2\ . \qquad (*)$$

Here, a is a constant.

(1) Transform the equation $(*)$ to a linear differential equation. The substitution $z=y^{-1}$ might be used.

(2) Solve the equation $(*)$ where $a=0$.

(3) Solve the equation $(*)$.

Problem 2

Consider the 3×3 matrix A.

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ -2 & -2 & 0 \end{pmatrix}$$

(1) Find the determinant $|A|$.

(2) Find the eigenvalues (λ_1, λ_2 and λ_3) and the matrix P satisfying $P^{-1}AP = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$.

Assume $\lambda_1 < \lambda_2 < \lambda_3$.

(3) Define $\vec{y}$ and $\vec{y'}$ as

$$\vec{y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \text{ and } \vec{y'} = \begin{pmatrix} \frac{dy_1}{dx} \\ \frac{dy_2}{dx} \\ \frac{dy_3}{dx} \end{pmatrix}.$$

y_1, y_2 and y_3 are the functions of x. Consider the system of differential equation expressed as $\vec{y'} = A\vec{y}$. Define $\vec{z} = \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix}$ satisfying $\vec{y} = P\vec{z}$ by using the matrix P in (2).

Obtain general solutions of z_1, z_2 and z_3.

(4) Obtain solutions of y_1, y_2 and y_3 in (3) satisfying the boundary condition that $y_1 = -2$, $y_2 = 3$ and $y_3 = 3$ at $x = 0$.

Problem 3

Consider a complex mapping function $w = f(z)$, which is denoted by two complex variables $z\ (= x + iy)$ and w, where x and y are two real variables and i is the imaginary unit. The complex function $f(z)$, which is analytic and consists of the real part $u(x, y)$ and the imaginary part $v(x, y)$, is given by:

$$f(z) = u(x, y) + iv(x, y).$$

Here, the two real functions, $u(x, y)$ and $v(x, y)$, satisfy the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

(1) Show that the real function $v(x, y)$ satisfies the Laplace's equation:

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0.$$

Answer the following questions when a boundary condition for $u(x, y)$ and the function form of $v(x, y)$ are given by

$$u(0, 0) = 0,$$
$$v(x, y) = xy.$$

(2) Find $u(x, y)$ and $f(z)$ by solving the Cauchy-Riemann equations.

(3) When the inverse mapping function is given by $z = f^{-1}(w)$, draw the curves on the z-plane, into which the following four straight-lines on the w-plane ($w = u + iv$) are mapped:

$$v = 0,\ v = 1,\ u = 0, \text{ and } u = 1.$$

The curves can be drawn within the range of $x \geq 0$ and $y \geq 0$.

(4) When an infinitesimal segment dw on the w-plane and a corresponding infinitesimal segment dz on the z-plane are defined, the relationship of $\frac{dw}{dz} = \frac{d\{f(z)\}}{dz}$ holds. Suppose that one straight line satisfying $v = \text{constant}$ and another straight line satisfying $u = \text{constant}$, which are orthogonal to each other on the w-plane, are inversely mapped into the curves on the z-plane. Explain that these curves on the z-plane are also orthogonal to each other, by using the relationship mentioned above.

(5) According to the result of (1), a group of curves satisfying $v(x, y) = \text{constant}$ on the z-plane is considered as equipotential lines in electromagnetism. According to the result of (4), a group of curves satisfying $u(x, y) = \text{constant}$ is considered as electric lines of force in electromagnetism. A vector $\boldsymbol{E}(x, y)$ acting along an electric line of force, which corresponds to an electric field vector, is given by,

$$\boldsymbol{E}(x, y) = -\text{grad}\ v(x, y).$$

Obtain the vector $\boldsymbol{E}(x, y)$ on the z-plane.

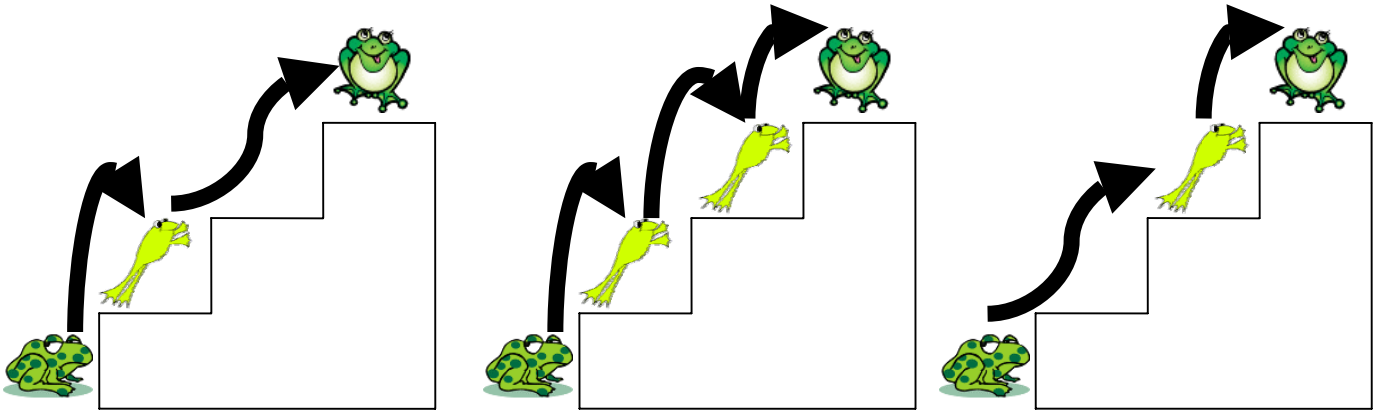
Problem 4

A frog has to climb an *n*-step stairs. Note that $n \geq 1$ and that the frog has to keep the following regulations:

(a) The frog is only allowed to climb up, and not to climb down.

(b) The frog may climb the stairs either one step or two steps at a time.

When the frog climbs a 3-step stairs, three ways are possible as shown below:



Simple notation : －－－

In the simple notation, climbing two steps at a time is represented by ＋, whereas climbing one step is represented by －. With these preparations, answer the following questions. Note that $S(n)$ represents the number of climbing ways for an *n*-step stairs.

(1) Enumerate all the climbing ways for $n = 4$ by using the simple notation.

(2) Calculate the values of $S(5)$ and $S(6)$.

(3) Derive the recurrence equation for $S(n)$.

(4) Prove the next equation with the binominal coefficient ${}_nC_k = \dfrac{n!}{(n-k)!k!}$:

$$S(n) = {}_nC_0 + {}_{n-1}C_1 + {}_{n-2}C_2 + \ldots + {}_{Trunc[(n+1)/2]}C_{Trunc[n/2]},$$

where *Trunc*[*x*] is a truncation function of rounding down a real value *x* to an integer value.

(5) Consider the right and upward summation of numbers in the Pascal triangle, as shown in the right figure. For example, the first six sums are $a(1) = 1$, $a(2) = 1$, $a(3) = 1+1$, $a(4) = 1+2$, $a(5) = 1+3+1$ and $a(6) = 1+4+3$. Derive the general term $a(n)$ of this sequence by using $S(n)$.

